MTH129 Introduction to AI for Mathematical Sciences (MS)

2025-2026 Academic Year
Semester 2

SMP AI Team

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Classification

Table of Contents

1 Hard vs. Soft Classification
2 Linear Discriminant Analysis (LDA)
3 Perceptron

Classification Paradigms: Hard vs. Soft

Fundamental Question

How should a classifier represent its predictions?

Soft Classification

- Outputs probability estimates
- $P(Y = k|\mathbf{x})$ for each class
- Quantifies prediction certainty
- Examples: Logistic regression, neural networks with softmax

Hard Classification

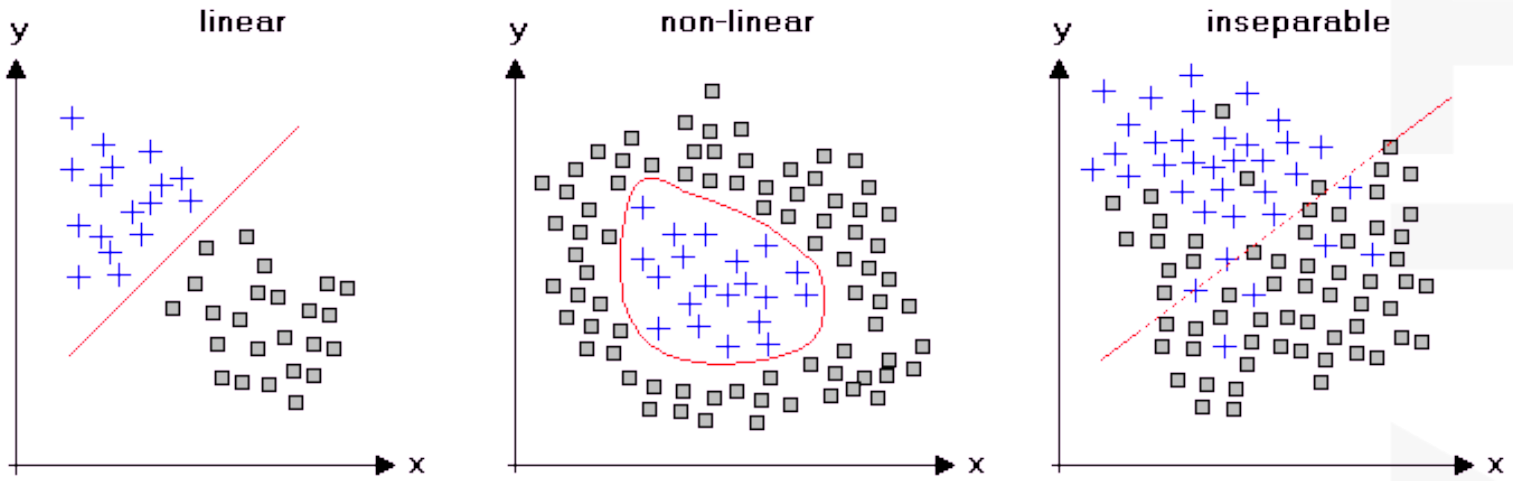
- Directly outputs class labels
- $\hat{y} = g(\mathbf{x})$
- No probability information
- Examples: Perceptron, LDA, SVM

Key Difference

Soft classifiers provide uncertainty quantification, while hard classifiers focus on decisive boundaries

Linear Separability: Core Concept

In the case of classification problems there are basically three cases to be distinguished: (1) data sets with linearly separable classes, (2) data with non-linearly separable classes, and (3) classes which cannot be separated at all.



Linear Separability: Core Concept

Definition: A dataset is linearly separable if there exists a hyperplane that perfectly separates classes

Mathematical Formulation

For binary classes $y \in \{-1, 1\}$:

$$\exists \mathbf{w}, b \quad \text{s.t.} \quad \begin{cases} \mathbf{w}^T \mathbf{x}_i + b > 0 & \forall i : y_i = 1 \\ \mathbf{w}^T \mathbf{x}_i + b < 0 & \forall i : y_i = -1 \end{cases}$$

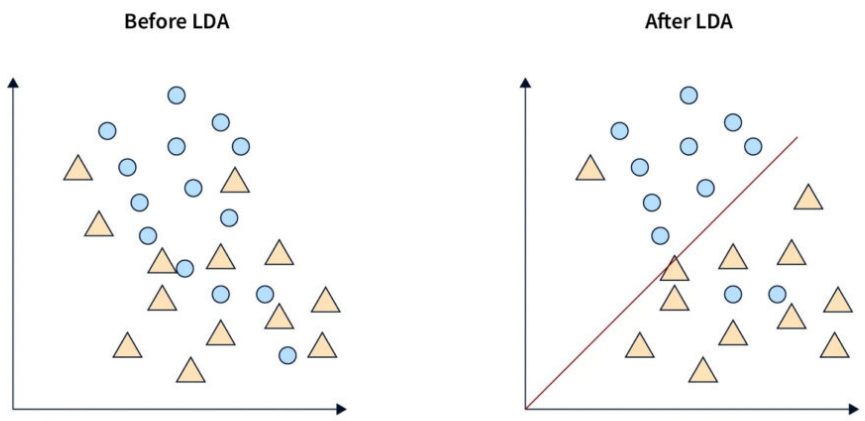
Test for linear separability

1. Perceptron converges in finite steps $\rightarrow$ separable
2. Convex hulls of classes don't intersect $\rightarrow$ separable
3. Linear program has feasible solution $\rightarrow$ separable

Linear Discriminant Analysis: Model Overview

Core idea: Find linear combinations of features that best separate classes

- Generative model: Models class-conditional densities
- Assumptions:
  - Gaussian class-conditional densities
  - Shared covariance matrix across classes
- Objective: Maximize between-class variance relative to within-class variance



Comparison with Logistic Regression

Aspect	**LDA**	**Logistic Regression**
Model type	Generative	Discriminative
Distribution	Assumes Gaussian	No distributional assumption
Decision	Linear boundary	Linear boundary
Output	Hard class	Probability + class
Covariance	Shared	Not modeled

LDA: Mathematical Formulation

Class-conditional densities:

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_k)\right)$$

Bayes' theorem:

$$P(Y=k|\mathbf{X}=\mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{l=1}^{K}\pi_l f_l(\mathbf{x})}$$

Log-odds for binary case:

$$\log\frac{P(Y=1|\mathbf{x})}{P(Y=0|\mathbf{x})} = \log\frac{\pi_1}{\pi_0} - \frac{1}{2}(\boldsymbol{\mu}_1+\boldsymbol{\mu}_0)^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0) + \mathbf{x}^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0)$$

Linear decision boundary

Set log-odds to zero: $\mathbf{w}^T\mathbf{x}+b=0$
where $\mathbf{w}=\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0)$, $b=-\frac{1}{2}(\boldsymbol{\mu}_1+\boldsymbol{\mu}_0)^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0)+\log\frac{\pi_1}{\pi_0}$

Parameter Estimation in LDA

From training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$:

1. **Prior probabilities:**

$$\hat{\pi}_k = \frac{n_k}{n}, \quad n_k = \sum_{i=1}^{n} \mathbb{I}(y_i = k)$$

2. **Class means:**

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{n_k} \sum_{i:y_i=k} \mathbf{x}_i$$

3. **Shared covariance matrix (unbiased):**

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^T$$

Parameter Estimation in LDA

Fisher's discriminant perspective:

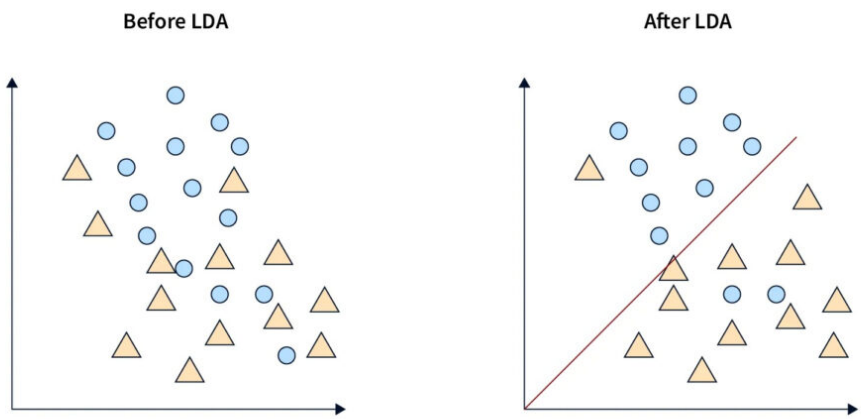
$$\max_{\mathbf{w}} J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}$$

where $\mathbf{S}_B = \sum_{k=1}^{K} n_k (\hat{\boldsymbol{\mu}}_k - \hat{\boldsymbol{\mu}})(\hat{\boldsymbol{\mu}}_k - \hat{\boldsymbol{\mu}})^T$, $\mathbf{S}_W = \sum_{k=1}^{K} \sum_{i:y_i=k} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^T$

Solution

$\mathbf{w}^*$ is the leading eigenvector of $\mathbf{S}_W^{-1}\mathbf{S}_B$ (for $K = 2$)

LDA: Geometric Interpretation



Key elements:

- Class centroids: μ_1, μ_2
- Within-class scatter: $\mathbf{S}_W$, Between-class scatter: $\mathbf{S}_B$
- Projection direction: $\mathbf{w}$

Decision boundary:

- Perpendicular to $\mathbf{w}$
- Passes through midpoint of centroids (when $\pi_1 = \pi_2$)

Optimal separation

LDA finds projection where:

- Classes are maximally separated
- Each class is tightly clustered

Perceptron: Model Overview

Historical context: Frank Rosenblatt (1957), first artificial neuron, the simplest AI!

Model definition:

$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^T\mathbf{x} + b)$$

where $\text{sign}(z) = \begin{cases} 1 & z > 0 \\ -1 & z \leq 0 \end{cases}$

Key properties:

- Hard classifier: Direct class prediction
- Linear decision boundary: $\mathbf{w}^T\mathbf{x} + b = 0$
- Online learning: Updates weights incrementally

Comparison with Logistic Regression

- Perceptron: Minimizes classification errors
- Logistic: Maximizes likelihood of probabilities
- Perceptron: No probability estimates
- Logistic: Sensitive to class imbalance

Perceptron Learning Algorithm

Loss function: 0-1 loss (non-differentiable)

$$L(\mathbf{w}) = \sum_{i=1}^{n} \mathbb{I}(y_i \neq \text{sign}(\mathbf{w}^T \mathbf{x}_i))$$

Perceptron criterion (surrogate loss):

$$J(\mathbf{w}) = -\sum_{i \in \mathcal{M}} y_i (\mathbf{w}^T \mathbf{x}_i)$$

where $\mathcal{M}$ is set of misclassified points

Update rule

For each misclassified $(\mathbf{x}_i, y_i)$:

$$\mathbf{w} \leftarrow \mathbf{w} + \eta y_i \mathbf{x}_i$$

- η: Learning rate (step size)
- Update only when misclassified

Perceptron: Interpretation

Weight update:

$$\mathbf{w} \leftarrow \mathbf{w} + \eta y_i \mathbf{x}_i$$

Case analysis:

- If $y_i = 1$ but $\hat{y} = -1$: $\mathbf{w}$ moves toward $\mathbf{x}_i$
- If $y_i = -1$ but $\hat{y} = 1$: $\mathbf{w}$ moves away from $\mathbf{x}_i$

Decision boundary:

- Hyperplane with normal $\mathbf{w}$
- Margin proportional to $\|\mathbf{w}\|$

Key limitation

Fails to converge if data not linearly separable (oscillates)

Perceptron vs. LDA vs. Logistic Regression

Feature	Perceptron	LDA	Logistic
Classification	Hard	Hard	Soft
Learning	Online	Batch	Batch
Probabilities	No	Yes (Bayesian)	Yes
Assumptions	None	Gaussian classes	None
Convergence	Finite (if separable)	Always	Always
Decision boundary	Linear	Linear	Linear
Sensitive to scaling	Yes	Yes	Yes
Solution uniqueness	No	Yes	Yes

When to use which?

- Perceptron: Simple, online learning; good baseline
- LDA: When Gaussian assumption holds; few data
- Logistic: Probability estimates needed; robust

Historical significance

Historical significance

Perceptron $\rightarrow$ Neural networks
LDA $\rightarrow$ Fisher discriminant analysis
Logistic $\rightarrow$ GLMs and deep learning